Threshold × k | Interventions and output length

Qwen3-1.7B student / Qwen3-14B teacher · 8 benchmarks × 25 configurations × 2 metrics = 400 values

MATH500

Baseline mean output tokens: Student 885.08 | Teacher 911.39; interventions 0

Mean interventions | k →

Threshold	1	4	8	16	32
2	9.71	8.67	8.00	7.46	6.09
2.5	6.29	5.72	5.16	4.79	4.50
3	4.41	3.83	3.69	3.46	3.22
3.5	3.08	2.69	2.76	2.65	2.52
4	2.15	2.12	2.06	1.83	1.85

Color scale: 1.83 – 9.71

Mean output tokens | k →

Threshold	1	4	8	16	32
2	842.03	824.53	837.39	879.40	808.50
2.5	829.13	807.28	830.83	809.57	832.38
3	841.23	823.16	814.26	808.41	854.73
3.5	866.49	834.26	818.37	838.94	868.77
4	822.84	885.79	869.41	840.68	854.41

Color scale: 807.28 – 885.79

OlympiadBench

Baseline mean output tokens: Student 1604.45 | Teacher 1524.58; interventions 0

Mean interventions | k →

Threshold	1	4	8	16	32
2	17.94	15.54	14.17	12.90	10.48
2.5	10.96	9.78	9.39	8.69	7.50
3	7.33	6.57	6.47	6.00	5.36
3.5	5.38	4.76	4.62	4.47	4.04
4	3.76	3.51	3.55	3.28	2.94

Color scale: 2.94 – 17.94

Mean output tokens | k →

Threshold	1	4	8	16	32
2	1510.33	1458.84	1482.34	1494.41	1409.54
2.5	1479.97	1482.64	1440.49	1441.71	1383.82
3	1482.96	1414.21	1456.42	1438.13	1377.94
3.5	1462.08	1443.61	1505.72	1387.82	1406.15
4	1523.21	1504.16	1515.12	1485.23	1379.40

Color scale: 1377.94 – 1523.21

BBH

Baseline mean output tokens: Student 479.75 | Teacher 279.43; interventions 0

Mean interventions | k →

Threshold	1	4	8	16	32
2	18.61	13.09	10.25	7.49	5.24
2.5	15.78	11.37	9.01	6.79	4.86
3	13.80	10.05	8.12	6.22	4.56
3.5	12.12	9.07	9.34	5.66	4.25
4	10.94	8.15	6.74	5.27	4.03

Color scale: 4.03 – 18.61

Mean output tokens | k →

Threshold	1	4	8	16	32
2	269.08	303.92	274.39	306.16	268.25
2.5	339.22	451.22	304.97	306.82	268.37
3	304.23	307.83	267.48	306.75	284.82
3.5	336.78	341.02	342.84	341.50	292.69
4	380.88	305.89	341.00	284.98	292.08

Color scale: 267.48 – 451.22

CRUXEval-O

Baseline mean output tokens: Student 402.01 | Teacher 115.12; interventions 0

Mean interventions | k →

Threshold	1	4	8	16	32
2	2.56	1.44	1.19	1.04	1.00
2.5	2.52	1.42	1.22	1.10	1.00
3	2.48	1.41	1.21	1.10	1.00
3.5	2.52	1.40	1.20	1.10	1.00
4	2.39	1.40	1.15	1.10	1.00

Color scale: 1.00 – 2.56

Mean output tokens | k →

Threshold	1	4	8	16	32
2	106.33	95.77	68.52	79.81	102.32
2.5	106.59	114.70	94.43	108.90	131.78
3	158.34	80.15	108.80	132.74	132.49
3.5	174.12	93.44	140.31	151.42	155.54
4	177.48	115.36	146.33	142.72	162.62

Color scale: 68.52 – 177.48

CRUXEval-I

Baseline mean output tokens: Student 349.06 | Teacher 18.23; interventions 0

Mean interventions | k →

Threshold	1	4	8	16	32
2	4.07	2.21	1.61	1.21	1.02
2.5	3.95	2.19	1.60	1.21	1.01
3	3.82	2.17	1.59	1.21	1.01
3.5	3.73	2.15	1.59	1.21	1.01
4	3.62	2.13	1.58	1.21	1.01

Color scale: 1.01 – 4.07

Mean output tokens | k →

Threshold	1	4	8	16	32
2	18.36	18.16	18.20	18.14	27.58
2.5	18.35	18.24	18.17	18.14	38.53
3	18.28	18.22	18.21	18.21	44.45
3.5	18.23	18.21	18.20	18.18	49.26
4	18.25	18.30	18.27	18.21	52.49

Color scale: 18.14 – 52.49

HumanEval+

Baseline mean output tokens: Student 199.18 | Teacher 177.63; interventions 0

Mean interventions | k →

Threshold	1	4	8	16	32
2	6.53	5.11	4.34	3.39	2.58
2.5	5.74	4.85	4.27	3.27	2.52
3	5.55	4.56	4.02	3.16	2.48
3.5	5.12	4.32	3.77	3.02	2.45
4	4.62	3.99	3.50	2.88	2.27

Color scale: 2.27 – 6.53

Mean output tokens | k →

Threshold	1	4	8	16	32
2	182.60	178.95	180.31	181.36	178.18
2.5	180.76	182.20	183.64	181.47	179.35
3	179.61	180.81	180.92	183.88	181.65
3.5	179.32	180.15	179.65	184.75	181.82
4	176.31	177.83	178.57	183.10	181.39

Color scale: 176.31 – 184.75

MBPP+

Baseline mean output tokens: Student 139.05 | Teacher 90.15; interventions 0

Mean interventions | k →

Threshold	1	4	8	16	32
2	6.13	4.55	3.79	2.76	2.01
2.5	5.80	4.45	3.55	2.70	1.95
3	5.19	4.15	3.43	2.61	1.92
3.5	4.92	4.02	3.24	2.52	1.88
4	4.69	3.91	3.11	2.49	1.81

Color scale: 1.81 – 6.13

Mean output tokens | k →

Threshold	1	4	8	16	32
2	91.89	92.10	91.52	89.21	91.13
2.5	94.10	97.02	91.12	90.38	91.35
3	94.48	94.77	93.68	92.11	93.89
3.5	96.02	96.07	93.10	93.36	96.48
4	97.59	99.31	94.14	94.97	95.24

Color scale: 89.21 – 99.31

LiveCodeBench

Baseline mean output tokens: Student 618.22 | Teacher 546.82; interventions 0

Mean interventions | k →

Threshold	1	4	8	16	32
2	28.93	21.90	17.83	13.34	9.69
2.5	22.47	17.67	14.25	11.78	8.91
3	18.18	14.95	12.67	9.36	7.81
3.5	14.44	11.93	10.41	8.57	6.55
4	11.63	10.30	8.83	7.35	6.07

Color scale: 6.07 – 28.93

Mean output tokens | k →

Threshold	1	4	8	16	32
2	567.96	572.71	524.41	516.69	536.02
2.5	593.76	561.63	554.13	555.95	552.02
3	563.92	609.34	580.23	513.47	570.83
3.5	580.63	531.71	554.88	562.46	609.21
4	620.26	582.14	563.57	574.39	580.33

Color scale: 513.47 – 620.26

Accuracy, interventions and length: independently scaled per benchmark. Teacher share: shared 0–100% scale. Darker = larger. Bold = best hybrid accuracy.
Teacher share excludes prompts / discarded drafts; it is not compute cost. Stopping rules and baseline generation differ. Full source: source.md.